

WAKAMATSU, Japan

WMO: 475700

Lat: 37.4884N

Lon: 139.9104E

Elev: 214

StdP: 98.78

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-4.9	-9.1	1.8	-3.4	-7.9	2.0	-2.4	8.2	-0.6	7.4	-0.5	1.4	300	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	33.5	24.6	32.0	24.1	30.4	23.4	25.4	31.5	24.7	30.3	24.2	29.0	3.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.1	27.7	23.2	18.5	27.3	22.7	17.9	26.8	79.1	31.6	76.5	30.5	74.2	29.1	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.4	5.6	-9.8	35.5	1.9	1.1	-11.2	36.3	-12.3	36.9	-13.4	37.6	-14.8	38.4		
			-10.1	26.5	1.9	0.5	-11.5	26.8	-12.6	27.1	-13.7	27.4	-15.1	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	-0.5	0.2	3.9	10.4	16.6	20.8	24.7	25.8	21.3	14.7	7.7	2.3
	DBStd	9.65	2.24	2.43	3.03	3.74	3.08	2.47	2.52	2.50	3.27	3.21	3.11	2.67
	HDD10.0	1155	324	276	191	39	1	0	0	0	3	83	238	
	HDD18.3	2802	582	509	448	238	71	7	0	10	121	321	496	
	CDD10.0	2025	0	0	1	51	204	325	456	489	338	147	13	0
	CDD18.3	632	0	0	0	1	16	82	197	231	98	7	0	0
	CDH23.3	4757	0	0	0	17	180	469	1472	2009	583	27	0	0
	CDH26.7	1658	0	0	0	2	32	102	538	818	163	4	0	0
Wind	WSAvg	2.0	2.3	2.3	2.4	2.4	2.3	2.0	1.7	1.7	1.7	1.7	1.8	2.1
Precipitation	PrecAvg	1336	85	63	77	79	90	124	198	165	157	121	84	96
	PrecMax	1656	203	99	143	140	165	218	388	518	284	348	159	204
	PrecMin	995	41	15	24	10	37	64	68	39	11	34	28	26
	PrecStd	179	39	20	27	36	35	42	72	100	61	78	34	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	9.7	18.3	25.8	29.6	31.1	34.7	35.7	32.7	26.3	18.4	12.3
		MCWB	5.1	5.7	10.4	14.7	17.8	20.9	25.0	25.2	23.7	20.1	13.6	9.7
	2%	DB	5.2	7.3	14.4	22.5	27.3	29.3	33.1	34.1	30.5	23.4	16.0	9.5
		MCWB	3.1	4.1	8.1	12.9	16.7	20.1	24.7	24.7	23.3	17.8	12.3	7.3
	5%	DB	3.9	5.6	11.6	20.1	25.3	27.7	31.6	32.7	28.5	21.7	14.5	7.9
		MCWB	2.0	3.0	6.9	12.1	16.0	19.5	24.0	24.6	22.2	16.9	11.5	5.9
	10%	DB	2.8	4.1	9.4	17.7	23.4	26.1	29.8	31.1	26.7	20.1	12.9	6.5
		MCWB	1.2	2.0	6.0	11.0	15.4	19.1	23.5	24.1	21.4	15.8	10.6	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	6.4	11.1	17.2	19.9	23.2	26.0	26.4	24.8	20.9	14.8	10.3
		MCDB	6.9	8.9	16.0	20.7	24.9	27.2	32.6	33.0	30.5	25.2	16.5	11.8
	2%	WB	3.5	4.6	9.2	14.5	18.4	22.1	25.2	25.6	23.8	19.1	13.2	7.7
		MCDB	4.9	6.6	12.9	20.0	23.9	25.7	31.2	31.9	28.9	22.1	15.2	9.0
	5%	WB	2.3	3.2	7.7	13.2	17.4	21.2	24.6	25.0	23.0	17.8	12.0	6.1
		MCDB	3.5	5.2	10.7	18.2	22.6	25.0	30.0	30.9	27.3	20.5	13.8	7.4
	10%	WB	1.4	2.1	6.1	12.0	16.5	20.4	24.0	24.5	22.2	16.6	10.7	4.9
		MCDB	2.6	3.9	9.0	16.4	21.5	24.3	28.6	29.7	25.6	19.4	12.6	6.2
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.7	8.8	11.3	11.4	9.2	8.4	9.2	8.8	9.1	7.9	5.7
		MCDBR	6.7	9.0	14.1	16.2	15.7	12.4	11.3	11.6	10.9	10.9	10.4	8.0
		MCWBR	4.8	6.2	8.5	7.8	6.5	4.8	4.2	4.0	4.6	5.5	6.6	5.9
	5% WB	MCDBR	6.3	8.6	12.7	13.1	11.7	9.0	9.9	10.4	9.8	8.8	8.8	7.6
		MCWBR	4.9	6.3	8.3	7.1	5.8	4.0	4.0	3.9	4.4	5.0	6.3	5.9
Clear-Sky Solar Irradiance	taub	0.403	0.453	0.533	0.583	0.598	0.617	0.621	0.613	0.526	0.470	0.426	0.403	
	taud	1.958	1.831	1.749	1.717	1.727	1.747	1.797	1.811	1.986	2.045	2.078	2.048	
	Ebn at Noon	732	743	725	720	718	704	697	689	728	726	709	704	
	Edh at Noon	136	176	211	230	232	227	215	207	165	141	121	118	
All-Sky Solar Radiation	RadAvg	1.80	2.61	3.60	4.45	5.14	4.85	4.48	4.58	3.74	2.89	2.05	1.60	
	RadStd	0.15	0.21	0.27	0.43	0.53	0.32	0.69	0.52	0.31	0.26	0.14	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page